Physics-Based Fusion of Sentinel-2 and Sentinel-3 for Higher Resolution Vegetation Monitoring

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***Abstract*— Monitoring vegetation growth, phenology, and health in agriculture requires very high spatial and spectral resolution sensors. Sentinel-2 is among the spaceborne sensors that tried to meet these requirements. However, due to physical constraints, few visible bands are present with relatively large spectral responses, which limits the leaf pigment content estimation. In this work, we propose to fuse Sentinel-2 bands with a Sentinel-3 image, which has a lower spatial resolution but contains several spectral narrower bands in both the visible and near-infrared (NIR) domains. The fusion procedure consists in sharpening the Sentinel-3 bands to match the Sentinel-2 spatial resolution leading to a higher spatial resolution Sentinel-3 image. The proposed fusion technique follows a physics-based approach based on the use of a radiative transfer model (RTM) for establishing a correspondence between Sentinel-2 and Sentinel-3 images. In greater details, the main spectrally pure constituent of each high-resolution Sentinel-2 pixel is identified for each pixel, and it is then related to the corresponding low-resolution Sentinel-3 pixel. Pure materials are the barycenters of clusters obtained by an unsupervised classification of the Sentinel-2 image. Sentinel-3 signatures are obtained using coarse resolution pixel matching with the Sentinel-2 image. The latter result is then corrected using radiative transfer modeling allowing the production of more realistic signatures. Validation was done using real Sentinel-2/Sentinel-3 images taken with a delay of only one day or less and by comparing the sharpened Sentinel-3 bands Oa04, Oa06, Oa08, and Oa17 with the corresponding Sentinel-2 bands B2, B3, B4, and B8A, as they are, respectively, spectrally close. For all the bands, the rms is lower than 0.007, and compared with the state-of-the-art techniques, it shows competitive results and robustness against scene heterogeneity. Besides, our findings prove that the retrieved Sentinel-3 signatures at full resolution (FR) are physically consistent and in good agreement with the Sentinel-2 data.**

***Index Terms*— Dictionary learning, image decomposition, image fusion, radiative transfer.**

I. Introduction

EARTH observation satellites are equipped with optical remote sensing systems designed to acquire images with different spatial and spectral resolution trade-offs defined by applications and technology limitations. For instance, to monitor urban development, it is, in general, important to have images with very high spatial resolution (decametric or submetric). These type of images are usually acquired by multispectral (MS) sensors (e.g., Pléiades, Skybox, and Worldview) and have few (below 20) large bandwidth spectral bands. For vegetation studies, such as monitoring crop growth, phenology, and yield, high spatial resolution sensors (decametric) are also necessary to distinguish between different crop fields. In terms of spectral resolution, it is necessary to have spectral bands sampling the critical intervals in the vegetation spectrum, such as the visible and red edge to detect pigment contents, the near-infrared (NIR) sensitive to leaf area index, and the shortwave infrared to detect water content [1]. Sentinel-2, Landsat-8, and Landsat-9 are the examples of MS satellites equipped with sensors with these characteristics. To study global effects, such as forest management, and global vegetation carbon uptake, only low spatial resolution is generally required, whereas high spectral resolution is necessary for precise retrieval of vegetation biophysical and biochemical properties and fluorescence measurements. One can cite the two European Spatial Agency (ESA) missions: the Sentinel-3 (operational) and fluorescence explorer (FLEX) (in preparation) equipped by the MS and hyperspectral (HS) sensors Ocean [2].

For some applications, it is useful to have, at the same time, high spatial, spectral, and temporal resolutions. These combinations could not be jointly ensured using the existing operational Earth observation satellites. For instance, to detect diseases on crops, it was recommended to use an HS sensor onboard aircraft for meeting the requirements of very high spatial resolution [3]. In order to overcome the technological limitations affecting spaceborne sensors, the fusion of multiresolution acquisitions revealed to be a suitable solution computationally providing both high spatial and spectral resolution images [4]. For instance, one can cite pansharpening, where the objective is to fuse a low spatial resolution MS with a high-resolution panchromatic (PAN) one. The result is a sharpened MS image at the same resolution as the PAN [5]. Following the pansharpening example, different techniques have been proposed to fuse HS with MS images. In fact, the HS acquisitions have a lower spatial resolution compared to MS images. Multiresolution image fusion techniques can be classified into four kinds; component substitution (CS) [6], multiresolution analysis (MRA) [7], deep learning-based techniques [5], and model-based techniques [8]. In the latter group, there are three main categories: Bayesian, unmixing, and sparse coding models [9].

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Another example of fusion happens in the spatial–temporal domain where the objective is to fuse high-spatial low-temporal resolution observations with low-spatial high-temporal resolution images, as most of the sensors are a complementary on spatial and temporal resolutions. The goal of the fusion is to obtain a both high-spatial and high-temporal resolution image time series. In this problem, it is possible to distinguish two kinds of fusion; spatial-weighting-based and spatial-unmixing-based. The spatial and temporal adaptive reflectance fusion model (STARFM) is an example of the first kind in which Landsat images at 30-m resolution and a revisit period of 16 days are fused with daily moderate-resolution imaging spectroradiometer (MODIS) images at a spatial resolution of 250–1000 m [10]. Each pixel of the predicted Landsat image is expressed as a linear combination over a window of the MODIS pixels at the same date comprising for a bias due to the resolution and central wavelength differences. The bias is derived at a previous date in which acquisitions from both sensors are available. The coefficients of the linear combination depend on the correspondence of pixels within the window. Fusion techniques based on spatial unmixing produce a classification map at high resolution and derives the corresponding class spectra of the spatially lower resolution image. This allows to predict the temporal variations of the high-resolution satellite image pixelwise without considering averaging over windows [11]. The latter technique benefits from the notion of virtual image pair (VIP), which allows for minimizing the estimation variance.

Spatial–spectral techniques are not limited to pansharpening and MS–HS fusion where the resolution ratio between high and low spatial resolution is two or four, but it is extended to the fusion of different sensors with a potentially greater difference in spatial and spectral resolutions. In this context, we can talk about downscaling techniques. For instance, in [12], it was proposed to fuse thermal infrared Spinning Enhanced Visible and InfraRed Imager (SEVIRI)/Meteosat data at low resolution (about 3 km) with optical Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) data at high spatial resolution (15 m). The result is a temperature image at the agriculture field scale. Fusion concerns also optical sensors in the visible solar reflective domains. In [13], it was proposed to downscale full-resolution (FR) Envisat Medium Resolution Imaging Spectrometer (MERIS) data (300-m resolution) with Landsat thematic mapper (TM; 25-m resolution). The result is a MERIS FR image at the same spatial resolution as Landsat TM. The method is based on a spatial-unmixing technique where a high-resolution classification map is produced allowing after that the prediction of the different MERIS class signatures. Later, this technique has been enhanced by using predefined spectral signatures in order to reduce uncertainty in the solution [14]. Recently, it was shown that it is possible to apply such a technique for Sentinel-3 downscaling using Sentinel-2 high spatial resolution image [15]. As four Sentinel-3 images share the same spectral properties as four from Sentinel-2, they are first downscaled. Then, they are used to downscale the other bands using the VIP approach and spatial-weighting-based or spatial-unmixing-based techniques [11]. In general, the downscaling processing is done over windows to deal with spatial variability. Such composition yields block artifacts appearing at the window borders. To overcome this drawback, it was proposed in [16] to add regularity terms that ensure that the class signature does not vary across windows.

In general, spatial-unmixing technique suffers from reflectance dynamic reduction, since the high-resolution product pixels belonging to the same class within the same block are given the same signature. Moreover, misregistration artifacts can strongly affect the class signatures, especially for underrepresented ones. Moreover, neither spatial-unmixing nor approach-based pansharpening takes into account the relationship between the high-resolution and low-resolution image signatures, which could yield to unrealistic signatures. Indeed, multiresolution images represent the same area leading to a physical relationship between the reflectances taken in the different bands. For instance, if the surveyed area contains high-density vegetation, red bands should have relatively low values, whereas high reflectance should be observed in red-edge and NIR bands. When studied areas are mainly agricultural fields and forests, vegetation radiative transfer models (RTMs) [17], [18] can be helpful in linking spectral signatures to vegetation properties. In this case, RTM inversion can allow the retrieval of the vegetation properties given the spectral signature from Sentinel-2 images; then, the same model can simulate the signature of Sentinel-3 properly taking into account the variation in the spectral acquisitions and the observation geometry. This way, Sentinel-3 signatures can be derived given the Sentinel-2 ones through a RTM. The use of RTM can be interpreted as a way to restrain the reconstructed spectral signatures to physically coherent solutions.

In this work, we propose an approach to fuse Sentinel-2 and Sentinel-3. We use the spatial-unmixing technique to derive the Sentinel-3 end-member signatures. We correct the latter given the signature retrieved using RTM. We produce each high-resolution Sentinel-3 pixel signature given its decomposition on end-members. The latter decomposition is obtained by dictionary learning on Sentinel-2.

In summary, the main contributions of this study are as follows:

1) the development of an approach to derive Sentinel-3 images at 10-m resolution by fusion with Sentinel-2;
2) the integration of an RTM inversion technique that allows deriving a priori of the Sentinel-3 surface reflectance given the Sentinel-2 corresponding reflectance spectra;
3) the definition of a technique to derive Sentinel-3 land occupation class signature given the Sentinel-2 one based on spatial constraints of Sentinel-3/Sentinel-2 coregistration and physical constraints regarding Sentinel-3 a priori.

The remainder of this article is organized as follows. In Section II, we introduce some background. In Section III, we detail the proposed framework. In Section IV, we present our experiments and results. Finally, we draw out some concluding remarks in Section V.

II. Theoretical Background

In this section, we will recall the physical and mathematical foundations of our approach.

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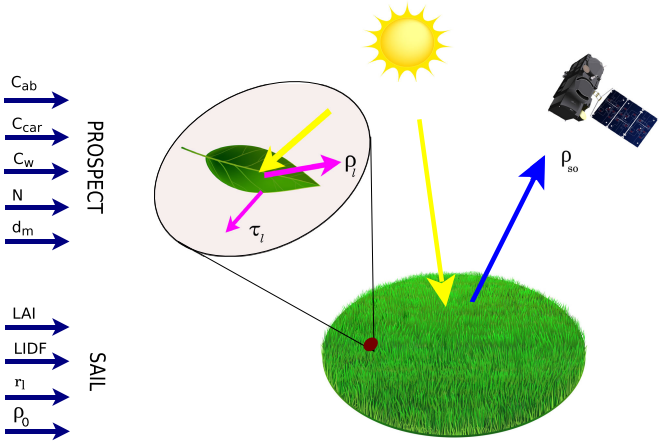


Fig. 1. PROSAIL diagram at the canopy level with a zoomed-in view over the leaf level.

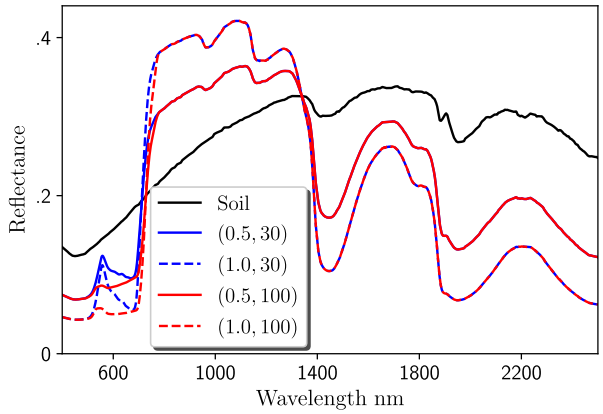


Fig. 2. PROSAIL reflectance spectrum simulation. $\Omega_s = (30°, 0°)$, $\Omega_o = (0°, 0°)$, $N = 1.2$, $C_{\text{car}} = 20\ \mu\text{g/cm}^2$, $C_{\text{w}} = 0.025$ cm, $d_{\text{m}} = 0\ \text{g/cm}^2$, $\bar{\theta}_l = 30°$, and $r_l = 0.1$. Values in parenthesis in legend are LAI and C_{ab}. Soil curve is the soil reflectance spectrum.

A. Radiative Transfer

RTMs simulate the interaction between light waves and different media [19]. RTMs are mainly used in remote sensing to simulate the reflectance of the Earth's surface taking into account the interaction between the different components. PROSAIL is among one of the well-known models [20]. It couples PROSPECT [1] and SAIL [17] at leaf and vegetation canopy scale, respectively. PROSPECT simulates the leaf hemispherical reflectance (ρ_l) and transmittance (τ_l). It takes into account the refraction index of the leaf, the structural parameter (N) that mimics the leaf cell layer number, and biochemical components, such as chlorophyll (C_{ab}), carotenoid (C_{car}), water contents (C_{w}), and dry matter (d_{m}). The chlorophyll and carotenoid pigments highly absorb the radiation in the visible domain, whereas water absorption is high in the shortwave infrared. The presence of dry matter impacts in all the NIR spectrum. For a given direction of the sun (Ω_s) and observation (Ω_o), SAIL simulates the canopy bidirectional reflectance (ρ_{so}) taking into account the leaf reflectance and transmittance as well as the vegetation biophysical properties, such as the leaf area index (LAI), the hot-spot effect (modeling the mutual-shading impact), parameterized by the ratio between the leaf radius and the vegetation (r_l), and the leaf inclination distribution function (LIDF), generally represented by the average leaf inclination ($\bar{\theta}_l$) [12], [18]. Finally, it is possible to include a characterization of a soil background in the simulation. The soil reflectance, ρ_0, is assumed in this case Lambertian. Fig. 1 presents the diagram of the PROSAIL model. Given the properties of the canopy and the vegetation SAIL simulates the radiative transfer and allows to predict the reflectance ρ_{so} as observed by the sensor. At the leaf level, SAIL uses the reflectance and the transmittance simulated by PROSPECT.

Fig. 2 shows an example of PROSAIL simulation of canopy reflectance in different cases given the soil signature. When LAI increases from 0.5 to 1, one can see that the reflectance decreases in the visible and the shortwave infrared domains and conversely increases in the NIR. Indeed, in the former case, when the vegetation cover rises, the leaf pigments (i.e., C_{ab} and C_{car}) and water (C_{w}) absorption effects increase consequently, which reduces the scattering and, therefore, the reflectance. Conversely, in the NIR, the leaf absorption is too low, and when the vegetation covers all of the surface, the soil absorption effect decreases, which yields with an increase in reflectance. The chlorophyll variation from 30 to 100 $\mu\text{g/cm}^2$ impacts the visible domain and also the red edge (the part of the spectrum of rapid increase of reflectance in the NIR domain).

RTMs are widely used for vegetation property retrieval. This is achieved by inversion using satellite data. One can cite the lookup table (LUT) inversion [21] as one of the most commonly used approached. LUT inversion consists of creating a large database by varying the possible properties and simulating for each combination of parameters the corresponding satellite observation. The inversion is done by comparing the real satellite observation to the simulated one and choosing the configuration of the biophysical properties giving the best fit.

B. Dictionary Learning

Given N vectors $\{x_i\}_{i=1,\ldots,N}$ with nonnegative entries of dimension M and a dictionary of D atoms $\{d_j\}_{j=1,\ldots,P}$, then each element x_i can be written as a linear combination of d_j, $j = 1, \ldots, P$

$$x_i = \sum_{j=1}^{P} w_{ij} d_j + \delta_i, \quad w_{ij} \geq 0 \tag{1}$$

where w_{ij} is the weight of the atom d_j in the composition of x_i and δ_i the decomposition error.

The corresponding matrix expression is

$$X = WD + \Delta \tag{2}$$

where $X \in \mathbb{R}_+^{N\times M}$, $\Delta \in \mathbb{R}^{N\times M}$, $D \in \mathbb{R}_+^{P\times M}$, and $W \in \mathbb{R}_+^{N\times P}$ are the set of the elements $\{x_i\}_{i=1,\ldots,N}$, $\{\delta_i\}_{i=1,\ldots,N}$, $\{d_j\}_{j=1,\ldots,P}$, and $\{w_{ij}\}_{i=1,\ldots,N;\, j=1,\ldots,P}$, respectively. The decomposition matrix W is, in general, unknown and can be found solving an optimization problem. Linear regression can be used in this case with the constraint that all its elements are nonnegative.

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To ensure suitable solutions, a sparsity penalty on each vector w_k is considered. Sparsity is commonly enforced by using the pseudonorm ℓ_0 or the norm ℓ_1. In our case, the ℓ_1 norm is chosen, as it ensures the problem convexity. Our problem writes as follows:

$$\hat{W} = \arg\min_{W} \left\{ \|X - WD\|_2^2 + \beta_{\ell_1} \|W\|_1 \right\}, \quad w_{ij} \geq 0 \quad (3)$$

where β_{ℓ_1} is a constant parameter, and $\|\cdot\|_1$ and $\|\cdot\|_2$ are the ℓ_1 and ℓ_2 norms, respectively.

III. Methodology

The Sentinel-2 and Sentinel-3 images are represented as $(S_2 = \{s_{2,i,j}\}_{i,j})$ and $(S_3 = \{s_{3,i,j}\}_{i,j})$, respectively. Their number of rows, columns, and bands is (N_2, M_2, P_2) and (N_3, M_3, P_3), respectively. As the resolution ratio between images is 30, therefore, $N_2 = 30N_3$ and $M_2 = 30M_3$. The result of the fusion process is a Sentinel-3 image at 10-m resolution, noted as $\hat{S}_3 = \{s_{3,i,j}\}_{i,j}$ of dimension (N_2, M_2, P_3).

Some assumptions on $\hat{S}_3$ are considered in the estimation. First, when $\hat{S}_3$ is downsampled to its original resolution, it should be as close as possible to S_3. Vegetation clusters are the same in $\hat{S}_3$ and S_2, as they represent the same scene. The Sentinel-3 cluster signatures can be deduced from the Sentinel-2 ones, assuming that the average cluster signature within each coarse resolution pixel is close to the cluster signature (averaged over all the image pixels) and that Sentinel-2 and Sentinel-3 signatures are compatible (i.e., for each cluster, the whole signature containing Sentinel-2 and Sentinel-3 wavelengths is a smooth and realistic curve). The high-resolution pixel compositions in terms of clusters is the same for both images, and it is derived from the Sentinel-2 signature.

Given the aforementioned assumptions, our processing chain is summarized in Fig. 3. First, the Sentinel-2 image [Fig. 3(a)] is classified using an unsupervised technique [Fig. 3(c)], and the cluster average values constitute the image dictionary atoms [Fig. 3(d)]. Then, using the relationship between Sentinel-2 and Sentinel-3 [Fig. 3(b)], it is possible to deduce the Sentinel-3 atoms [Fig. 3(e)]. Sentinel-2 pixel decomposition as the functions of atoms is computed [Fig. 3(f)]. As it is assumed the same for Sentinel-3 and given the atoms of the latter, it is possible to reconstruct $\hat{S}_3$ at 10-m resolution [Fig. 3(g)].

A. Dictionary Learning

Due to the Sentinel-3 low resolution, it is impractical to detect the signature of the different pure classes of occupation, such as soil and vegetation, that will constitute our dictionary. For that, we will use as a proxy the Sentinel-2 image. We start with the "pure material" signatures (i.e., spectra of materials fully spatially resolved) of Sentinel-2 to deduce after that the corresponding Sentinel-3 ones.

Pixelwise classification done on the Sentinel-2 image at 10-m resolution identifies the "pure materials" (i.e., end-members) that can be, for instance, soil, road, building, different kinds of crops, and so on. Soil and vegetation signatures in the domain (400, 2500 nm) are well studied in the literature, and given their values on some wavelengths, it is possible

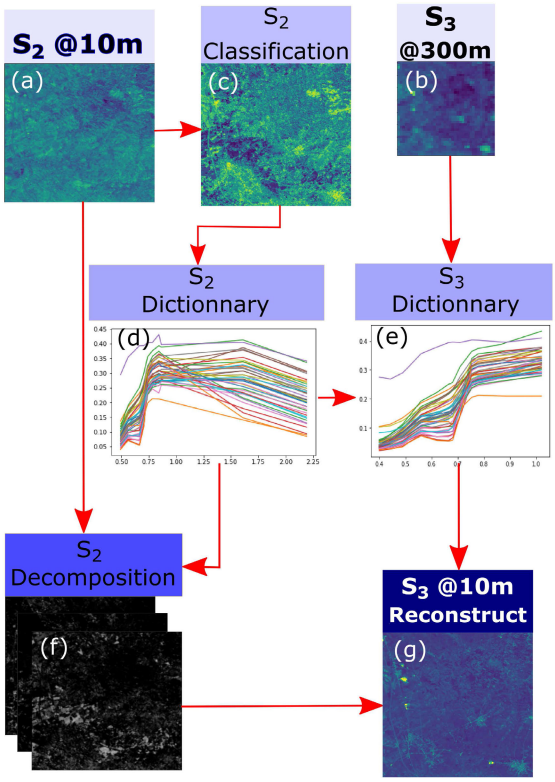


Fig. 3. Flowchart of the Sentinel-3 and Sentinel-2 fusion technique steps. (a) S_2@10 m. (b) S_3@300 m. (c) S_2 classification. (d) S_2 dictionary. (e) S_3 dictionary. (f) S_2 decomposition. (g) S_3@10-m reconstruct.

to extrapolate to the entire spectrum. The Gaussian mixture model (GMM) is used to do the unsupervised classification. This technique has shown its effectiveness in many satellite image applications [22]. The GMM is a probabilistic model that assumes that the data points (in our case the different Sentinel-2 pixel signatures) belong to a given number of clusters following a Gaussian distribution. The GMM is based on the expectation–maximization algorithm for fitting mixture-of-Gaussian models. The classification image is called $X = \{x_{ij}\}_{i,j}$. It has the same spatial dimensions as S_2 and $x_{ij} \in \{1, \ldots, K\}$, with K the number of clusters.

The Sentinel-2 dictionary $E_2 = \{e_{2,k}\}_k$ is formed by the center of these clusters. In the following, we will try to find the corresponding Sentinel-3 dictionary $E_3 = \{e_{3,k}\}_k$. In this case, $e_{2,k}$ and $e_{3,k}$ are the vectors of dimensions P_2 and P_3, respectively.

1) Image Registration: The Sentinel-3 image, which is initially in swath geometry, is projected into the same coordinate system as Sentinel-2 in order to get aligned pixels. As the resolution ratio is 30, each Sentinel-3 pixel corresponds ideally to 30×30 pixels of Sentinel-2. In terms of classification, the S_3 pixels contain mixed materials. Given the classification X at 10-m resolution, it is possible to define for each S_3 pixel its components in terms of clusters $(\Pi = \{\pi_{ijk}\}_{i,j,k})$. Π dimension is (N_3, M_3, K). $\pi_{ijk} \in [0, 1]$ represents the fraction of the pixel (i, j) that belongs to the cluster k

$$\sum_{k=1}^{K} \pi_{ijk} = 1. \quad (4)$$

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Then, it is possible to write

$$\sum_{k=1}^{K} \pi_{ijk} e_{3,k} = s_{3,i,j} + \epsilon_3 \tag{5}$$

where ϵ_3 is the noise that represents the uncertainty in this equation that it is due to the misalignment between images and the Sentinel-3 point spread function modeling the reduced spatial resolution.

The expression given by (5) can be seen as the data attachment term that depends directly on Sentinel-3. Then, it is possible to derive E_3 by solving a least square problem

$$\hat{E}_3 = \arg\min_{E_3} \sum_{i,j} \left\| \sum_{k=1}^{K} \pi_{ijk} e_{3,k} - s_{3,i,j} \right\|_2^2. \tag{6}$$

The solution of (6) is analytically given by

$$\hat{E}_3 = ({}^t\boldsymbol{\pi}\boldsymbol{\pi})^{-1\,t}\boldsymbol{\pi}\boldsymbol{s}_3 \tag{7}$$

where $\boldsymbol{\pi}$ and $\boldsymbol{s}_3$ of size (N_3M_3, K) and (N_3M_3, P_3) are deduced, respectively, from Π and S_3 by stacking the columns on top of one another.

2) Physical Constraint: The dictionary comprising the Sentinel-2 signatures can be split into a group containing vegetation canopy spectra ($E_2^{(v)} = \{e_{2,k}^{(v)}\}_{k=1,\ldots,K_v}$) and one the background materials ($E_2^{(0)} = \{e_{2,k}^{(0)}\}_{k=1,\ldots,K_0}$) (e.g., soil, road, and build-up areas). The distinction between vegetation and the other classes is a relatively easy task, and here, it is done by using the normalized difference vegetation index (NDVI). Indeed, vegetated areas exhibit high values of NDVI. In particular, a threshold of 0.2 was empirically fixed. Pixels with highest NDVI value than this threshold are assumed containing vegetation.

Similar to Sentinel-2, the Sentinel-3 dictionary is divided up into $E_3^{(0)} = \{e_{3,k}^{(0)}\}_{k=1,\ldots,K_0}$ and $E_3^{(v)} = \{e_{3,k}^{(v)}\}_{k=1,\ldots,K_v}$. These two groups are processed separately.

Given the Sentinel-2 signature of the background $E_2^{(0)}$, it is possible to predict the whole signature by linear interpolation. Indeed, in general, minerals, rocks, and soils do not present any specific absorption bands in the Sentinel-2/Sentinel-3 bands in the visible and NIR domains [23]. The vegetation signature is not easily predicted, since it presents high absorption bands mainly in the red spectral region, which makes interpolation highly imprecise. To overcome this difficulty, we will base our derivation on radiative transfer modeling. Indeed, given the Sentinel-2 signature of a vegetation canopy, it is possible to retrieve its biophysical property set, and then, with the latter, one can reconstruct the whole spectrum and, particularly, the Sentinel-3 signature. The retrieval is done in our case by inversion of the RTM PROSAIL. LUT is adopted as the inversion technique [24].

The reflectance of a given pixel (ρ) is modeled as a mixture between background materials (ρ_0) and a vegetation canopy (ρ_v)

$$\rho = p \times \rho_0 + (1-p) \times \rho_v(\Omega_{s,2}, \Omega_{o,2}, \rho_0, \mathrm{LAI}, \bar{\theta}_l, r_l, C_{\mathrm{ab}}, C_{car}, C_{\mathrm{w}}, N, d_{\mathrm{m}}) \tag{8}$$

where $p \in [0, 1]$ the mixing parameter, and $\Omega_{s,2}$ and $\Omega_{o,2}$ are the Sentinel-2 sun and sensor directions, respectively.

TABLE I

SIMULATION DATABASE VARIABLE VARIATION. WHEN THE UNIT IS NOT DECLARED, IT CORRESPONDS TO A DIMENSIONLESS VARIABLE. k_0 IS THE INDEX OF THE USED BACKGROUND

Variable	min	max	step
k_0	1	K_0	1
α_0	0.8	1.2	0.1
p	0	1	0.1
LAI (m^2/m^2)	0.25	3	0.25
$\bar{\theta}_l$ (degree)	20	70	10
r_l (m/m)	0.1	0.3	0.1
C_{ab} $(\mu g/cm^2)$	0	90	10
C_{car} $(\mu g/cm^2)$	0	40	10
C_{w} (cm)	0	0.06	0.005
N	1	3	0.5
d_{m} (g/cm^2)	0	0.02	0.05

ρ_0 represents the background reflectance and is assumed to be one within the set $E_2^{(0)}$. However, due to differences in moisture and surface roughness, spectral variations can be expected. Spectral variability can be modeled by

$$\rho_0 \in \left\{ \alpha_0 \times e_{2,k}^{(0)}, k \in \{1, \ldots, K_0\} \right\} \tag{9}$$

with α_0 a positive scalar value close to 1.

The variable variation domains are presented in Table I. After the creation of the database, the LUT technique is applied to retrieve the set of the properties relative to each element $e_{2,k}^{(v)} \in E_2^{(v)}$. Given these properties and the scene directions $(\Omega_{s,3}, \Omega_{o,3})$ relative to Sentinel-3, it is possible to simulate the reflectance $e_{3,k}^{(v)} \in E_3^{(v)}$.

The obtained dictionary using model inversion is called $E_3^{(\mathrm{inv})} = \{e_{3,k}^{(\mathrm{inv})}\}_{k=1,\ldots,K}$ in the following. This dictionary can be viewed as an a priori estimation of E_3. Then, linking data attachment [see (5)] to this a priori in a Bayesian formulation [12], it is possible to write

$$\hat{E}_3 = \arg\min_{E_3} \sum_{i,j} \left\| \sum_{k=1}^{K} \pi_{ijk} e_{3,k} - s_{3,i,j} \right\|_2^2 + \beta_{\mathrm{inv}} \sum_{k=1}^{K} \left\| e_{3,k} - e_{3,k}^{(\mathrm{inv})} \right\|_2^2 \tag{10}$$

where β_{inv} is the weighting parameter that controls the strength of the a priori compared to the data attachment.

The solution of this problem is given by

$$\hat{E}_3 = ({}^t\boldsymbol{\pi}\boldsymbol{\pi} + \beta_{\mathrm{inv}} I_K)^{-1} \left({}^t\boldsymbol{\pi}\boldsymbol{s}_3 + \beta_{\mathrm{inv}} E_3^{(\mathrm{inv})} \right) \tag{11}$$

where I_K is the identity matrix of size (K, K).

B. 10-m Sentinel-3 Construction

As the Sentinel-3 dictionary is known, to derive the Sentinel-3 image at 10-m resolution ($\hat{S}_3$), we need only to know the decomposition of these pixels [$W = \{w_{i,j}\}_{i,j}$ of size (N_2, M_2, K)].

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1) Sparse Decomposition Learning: As the decomposition is the same for Sentinel-3 and Sentinel-2, it is possible to learn the decomposition using the latter satellite data. For each pixel $(i, j) \in 1, \ldots, N_2 \times 1, \ldots, M_2$, the problem is written using the formulation given by (3)

$$\hat{w}_{ij} = \arg\min_{w_{ij}} \left\{ \|s_{2,i,j} - w_{ij} E_2\|_2^2 + \beta_{\ell_1} \|w_{ij}\|_1 \right\} \quad w_{ij}[k] \geq 0,\ k = 1, \ldots, K. \tag{12}$$

To solve (12), we use the least absolute shrinkage and selection operator (LASSO) algorithm [25] which is well adapted to such an optimization. Once solved, (12) allows to deduce $\hat{W}_{ij}$ and then construct the first version of the fusion product $\hat{S}_3^{(0)} = \{\hat{s}_{3,i,j}^{(0)}\}_{i,j}$ as follows:

$$\hat{s}_{3,i,j}^{(0)} = \hat{w}_{ij} E_3. \tag{13}$$

2) Bias Correction: Equation (14) can be applied for Sentinel-2, too

$$\hat{s}_{2,i,j}^{(0)} = \hat{w}_{ij} E_2. \tag{14}$$

$\hat{s}_{2,i,j}^{(0)}$ should be as close as possible to $s_{2,i,j}$. The difference between them is a reconstruction bias. Besides, if S_2 is not well retrieved, the fusion product at 10-m resolution will be inevitably affected. To overcome this issue, we propose to take into account the S_2 reconstruction bias in the S_3 10-m construction.

The S_2 bias reconstruction matrix $\Delta_2 = \{\delta_{2,i,j}\}$ of dimension (M_2, N_2, P_2) is defined as follows:

$$\delta_{2,i,j} = s_{2,i,j} - \hat{s}_{2,i,j}^{(0)}. \tag{15}$$

As a bias, $\delta_{2,i,j}$ can be assumed to vary slowly in the spectral domain. Therefore, it is possible to estimate it for the Sentinel-3 bands by simple linear interpolation. Exception is raised for the Sentinel-3 bands outside the Sentinel-2 band range. In this case, the nearest neighbor extrapolation is applied. The Sentinel-3 bias will be called $\Delta_3 = \{\delta_{3,i,j}\}_{i,j}$ of dimension (M_2, N_2, P_3). The final fusion product $\hat{S}_3$ is given by

$$\hat{S}_3 = \hat{S}_3^{(0)} + \Delta_3. \tag{16}$$

IV. Experimental Results

A. Date Presentation

1) Study Areas: Fusion of Sentinel-2 and Sentinel-3 leads to both high spectral and spatial resolutions, which is highly useful in monitoring vegetation canopies. In order to validate our work, we propose to study two contrasted canopies in Tunisia and another in Estonia. Fig. 4 shows the different areas: Zarmedine and Dar Chichou in Tunisia and Jarvselja in Estonia. Zaremedine is an agricultural area under a semiarid climate in the center of Tunisia. It is characterized by orchards of olive and almond trees. The second canopy is a mixture of forested and agricultural areas under a temperate climate. It is characterized by citrus orchards with agricultural fields of large fields. The forest is composed mainly by Aleppo pine. Jarvselja constitutes an example of a hemiboreal area and contains the training and experimental forestry district for the Estonian University of Life Sciences that served as a test for different projects, such as Valery [26], for missions such as Compact High Resolution Imaging Spectrometer (CHRIS)/Project for On-Board Autonomy (PROBA) [27]. There are different forest tree species present, such as pine, birch, and spruce. The studied area contains also some agricultural fields.

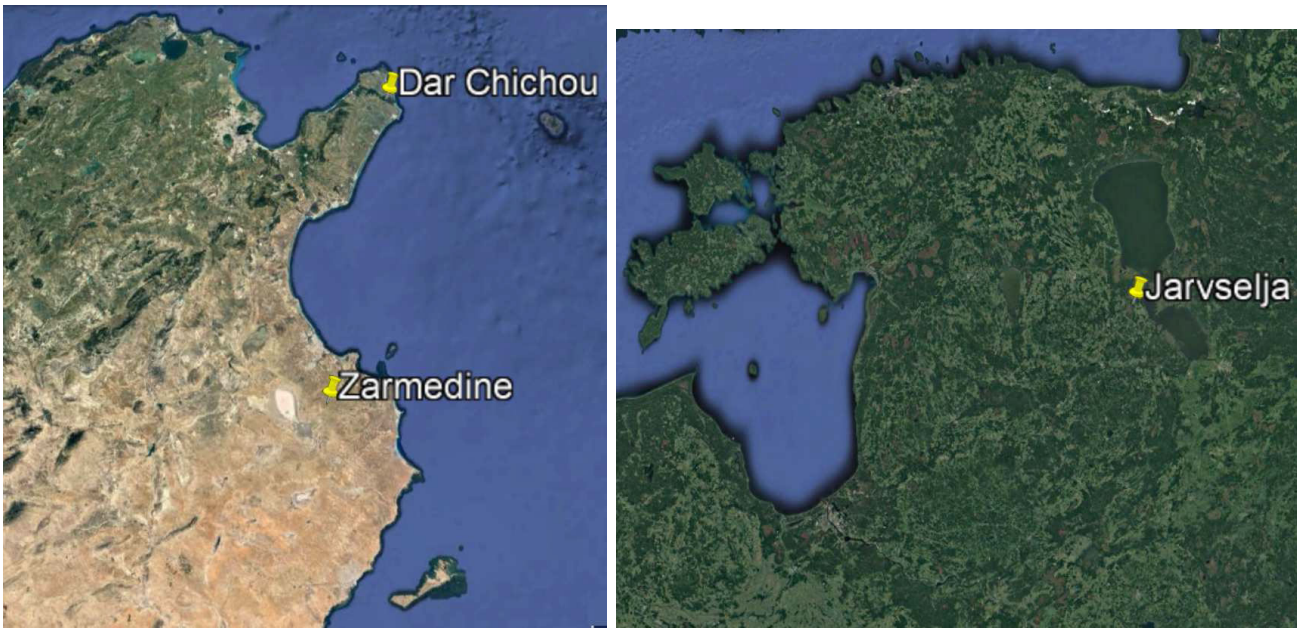
2) Remote Sensing Data: Sentinel-2 spectral and spatial resolutions are drawn in Table II. Spatial resolution is 10, 20, and 60 m according to the band. Those at 60 m are required for atmospheric corrections, as they are centered in the atmospheric absorption bands. The 10- and 20-m resolutions are useful for urban and vegetation study. In our work, we used the generalized Laplacian pyramid (GLP) technique to sharpen the 20-m image to 10 m [28]. Sentinel-3 properties are given in Table III. It is characterized by 21 narrow bands at 300-m spatial resolution. Some of these bands (i.e., Oa13, Oa14, Oa15, Oa19, and Oa20) are in the atmospheric absorption bands due to aerosol, O_2, and so on. Thus, only the 16 remained bands are useful for land surface reflectance measurements.

In terms of atmospheric correction, the bottom of atmosphere (BOA) reflectance product of Sentinel-2 is provided by the Thematic Center for Continental Surfaces (THEIA) [29]. For Sentinel-3, images are corrected using the software image correction for atmospheric effects (iCOR) [30]. Fig. 5 shows the spectra of Sentinel-3 after removing those in the atmosphere absorption domains and the Sentinel-2 bands at 10 and 20 m that belong to the same domain as Sentinel-3. It is clear here that Sentinel-3 has a higher spectral resolution, which can be helpful for vegetation biophysical property retrieval. Nevertheless, Sentinel-2 has many bands in the same domain, and those at 10 m are spectrally broad but at 20 m, they have comparable bandwidth to Sentinel-3. It is clear that if the canopy signature is known using Sentinel-2 observation, it will impose necessarily constraints on the Sentinel-3 signature.

To ensure the correspondence between Sentinel-2 and Sentinel-3 images, it is important to have the same geometry (i.e., the same sun and sensor orientations) for ensuring the same observation perspective and the same shadowing effect.

Figs. 6 and 7 show the original Sentinel-3 and Sentinel-2 images, respectively. For the Tunisian areas, Sentinel-2 images are almost taken simultaneously, as the satellite flies over the two areas successively. They are one day earlier than Sentinel-3. We assume that they are almost taken simultaneously, as the delay is too short for major land-cover changes. For the Estonian area, both satellite images are taken on the same day. Figs. 8 and 9 show the zoomed-in view over our two Tunisian areas in Zarmedine and Dar Chichou. Table IV summarizes the different information about the studied areas. Due to the highest rainfall in Cap Bon, Area 2 vegetation cover is the highest, and the corresponding visible image is the darkest. Indeed, we recall that vegetation pigments absorb visible light due to the photosynthetic activity. In particular, due to the tree density in the forest, the corresponding part in the image is darker than the other vegetation fields. Fig. 10 shows the zoomed-in view over the Jarvselja area. It is highly darker than the Tunisian areas, since it corresponds to a hemiboreal area with high tree density.

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(a) (b)

Fig. 4. Study areas in Tunisia (north Africa) and Estonia (northeast Europe). (a) Tunisia: Zarmedine (10°40′E, 35°30′N) in Monastir and Dar Chichou (11°5′E, 36°55′N) in Cap Bon region. (b) Estonia: Jarvselja (27°18′45″E, 58°16′04″N).

TABLE II
SENTINEL-2 BAND FEATURES

Band	B1	B2	B3	B4	B5	B6	B7	B8	B8a	B9	B10	B11	B12
Wavelength (nm)	442.7	492.4	559.8	664.6	704.1	740.5	782.8	832.8	864.7	945.1	1373.5	1613.7	2202.4
Bandwidth (nm)	21	66	36	31	15	15	20	106	21	20	31	91	175
Resolution (m)	60	10	103	10	20	20	20	10	20	60	60	20	20

TABLE III
SENTINEL-3 BAND FEATURES. THE SPATIAL RESOLUTION OF ALL THE BANDS IS 300 m

Band	Oa01	Oa02	Oa03	Oa04	Oa05	Oa06	Oa07	Oa08	Oa09	Oa10	Oa11
Wavelength (nm)	400	412.5	442.5	490	510	560	620	665	673.75	681.25	708.75
Bandwidth (nm)	15	10	10	10	10	10	10	10	7.5	7.5	10
Band	Oa12	Oa13	Oa14	Oa15	Oa16	Oa17	Oa18	Oa19	Oa20	Oa21	
Wavelength (nm)	753.75	761.25	764.375	767.5	778.75	865	885	900	940	1020	
Bandwidth (nm)	7.5	2.5	3.75	2.5	15	20	10	10	20	40	

TABLE IV
STUDY AREA INFORMATION SUMMARY

Area	S2 date	S3 date	S2 time	S3 time	S2 size	S3 size
Monastir	12/31/2020	01/01/2021	10:10	09:44	1194×1194	40×40
Dar Chichou	12/31/2020	01/01/2021	10:13	09:44	594×812	20×26
Jarvselja	06/18/2021	06/18/2021	09:40	08:43	1253×945	42×32

The comparison between Sentinel-2 and Sentinel-3 shows roughly the same areas but with different resolutions, as Sentinel-3 has a lower spatial resolution. For the Tunisian areas, Sentinel-2 images are acquired around 27 min later than those of Sentinel-3. This implies a slight difference in terms of solar zenith angle (θ_s), as it decreases in the morning. Nevertheless, the difference is relatively small ($\approx$2°). In terms of sensor inclination, we also tried to obtain close results. Thus, as Sentinel-2 observes on the nadir for almost all the scene and the Sentinel-3 field of view is too wide (68.5°), we carefully chose Sentinel-3, such that our study areas are observed close to nadir. In our case, the Sentinel-3 zenith angle is around 6°. Therefore, as Sentinel-2 and Sentinel-3 are almost within the same geometry, it is possible to validate the fusion result. For the Estonian area, Sentinel-3 was acquired almost 1 h earlier than Sentinel-2, and the zenith angle difference is $\approx$ 3°. In terms of sensor inclination, there is also a difference of $\approx$ 16°. These differences in acquisition

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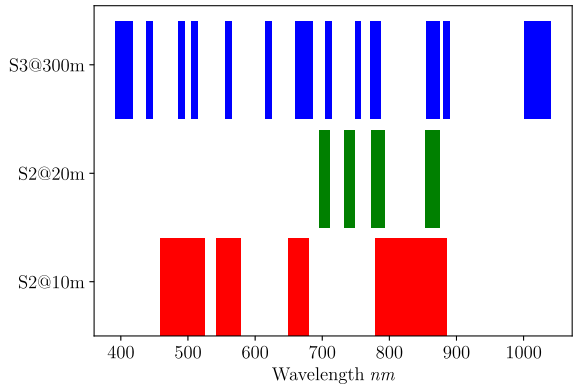


Fig. 5. Spectra of Sentinel-2 and Sentinel-3. S3@300 m are all the bands of Sentinel-3 except those in atmosphere absorption bands, S2@10 m are all the Sentinel-2 bands at 10 m-resolution, and S2@20 m are those at 20-m resolution and belonging to the same domain at Sentinel-3 bands.

geometry could make the Sentinel-2 and Sentinel-3 images relatively different. However, based on PROSAIL simulations, we have found out that the reflectance average bias is around 0.004 with this angular difference. We consider that this an error is acceptable for optical satellite image remote sensing applications [22].

B. Validation at FR

In order to validate our approach at FR, we need reference images at the target resolution. The reference should be taken with exactly the same parameters as the original Sentinel-3 image but at higher spatial resolution. In particular, from a geometrical point of view, the reference should be taken at the same time as Sentinel-3 to ensure the same sunlit direction and the same atmospheric conditions. In addition, the sensor should have the same orientation as the original image to ensure the same perspective. From a spectral point of view, the images should be taken within the same spectral bands. As such a reference image is not available, our conditions must be relaxed in order to find an approximate reference. We will assume that from a spectral point of view, B02, B03, B04, and B08A approximate Oa04, Oa06, Oa08, and Oa17, respectively [15]. Geometrically, the sun and sensor orientation differences between the two sensors should not have a significant impact. The weather conditions were suitable for satellite image acquisition in all the considered days limiting the presence of possible artifacts due to atmospheric correction. Given the aforementioned arguments, the fusion product Oa04, Oa06, Oa08, and Oa17 will be validated by comparison with B02, B03, B04, and B08A, respectively. Nevertheless, unlike Oa04, Oa06, and Oa08 bands, B02, B03, and B04 bands have wider spectral integration intervals. This difference of spectral width is a limitation of the comparison results. This is not the case for B08A, which fits well with Oa17. The validation is, therefore, much more meaningful for the B08A–Oa17 couple.

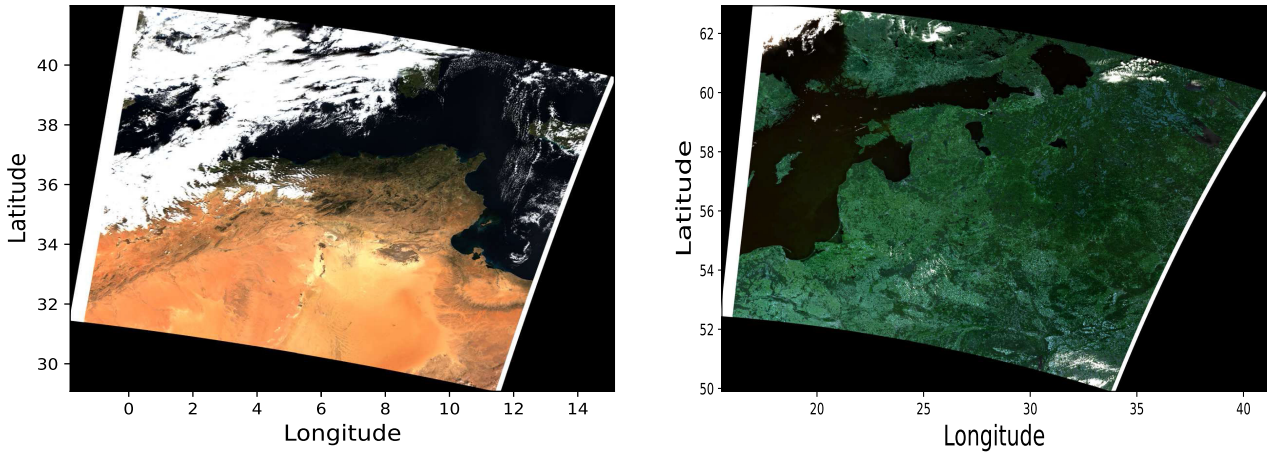
In the following, our proposed image fusion technique will be called fusion for high spatial and spectral resolution (FHSSR). The version of the method that does not correct the Sentinel-3 end-member spectra based on the radiative transfer modeling will be called FHSSR0. Comparing them will show the importance of considering the RTM.

Our model hyperparameters are the number of clusters K and the a priori multiplier β_{inv}. For K, we found out that a value around 30 or higher is enough to allow the distinction between the occupation classes and get acceptable fusion results. Taking further greater values for K does not have a significant enhancement of the results but can highly increase the problem computational complexity. For this reason, we have fixed $K = 40$ as a reasonable trade-off. Regarding β_{inv}, values between 5 and 100 with a step equals 5 have been tested. We found that best results are obtained around 20 and 40. For that, we chose $\beta_{\text{inv}} = 30$.

Our model is compared with different models of HS–MS image fusion. In particular, we compare with the Gram–Schmidt adaptive (GSA) [6] and smoothing filter-based intensity modulation (SFIM) techniques [31], which belong to the CS methods; the GLP [28] and the coupled multiresolution decomposition (CMD) technique [7], which belong to the MRA family; HS superresolution (HySure) and fast fusion based on Sylvester equation (FUSE) [32] belonging to the Bayesian framework; the coupled nonnegative matrix factorization (CNMF) [33] and the Lanaras's technique (ICCV'15) [34], which belong to the unmixing method; and, finally, the Akhtar's technique (ECCV'14) [35], which belong to the sparse coding-based approaches. In addition to the aforementioned methods, we will consider as a baseline the technique using spatial-unmixing described in [13]. We implemented it as suggested in the original article with blocks of size 7×7 pixels and classification with 25 clusters. All methods have been extensively studied in [5] and showed the best fusion performance for several datasets.

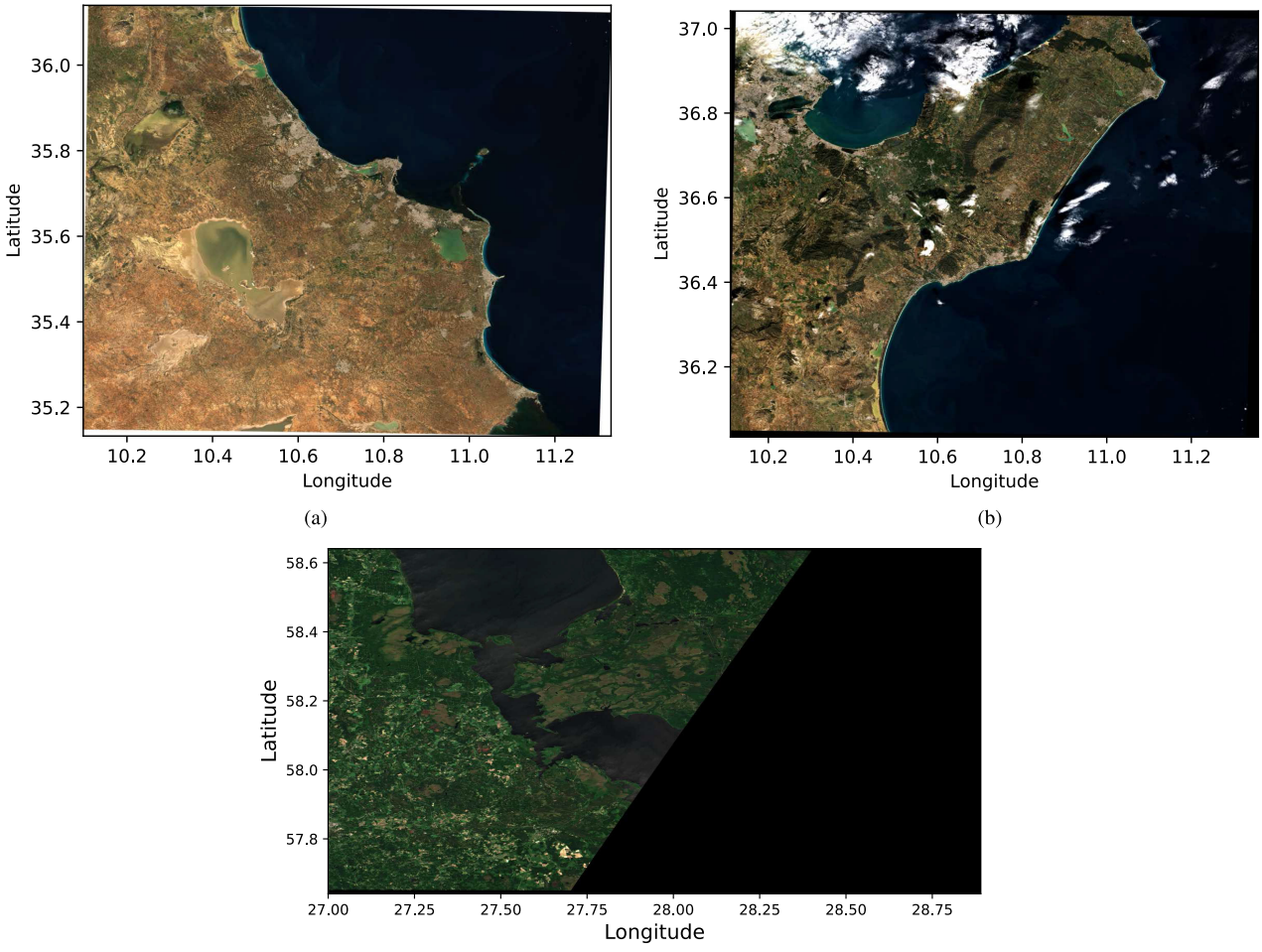
1) Quantitative Evaluation: The quantitative assessment is based on well-known quality metrics. In particular, for single band comparison, the root-mean-square error (RMSE) is used. Metrics for multibands are those conventionally used in the pansharpening literature. There are indices that measure spatial distortion, such as the relative dimensionless global error in synthesis (ERGAS) [36] and the peak-signal-to-noise ratio (PSNR). Spectral distortion is measured by the spectral angle mapper (SAM) [37]. In addition, we used the $Q2^n$, which is a generalization of the universal image quality index (UIQI) to MS and HS images, which was proposed to measure both the spatial and spectral distortions [38]. Tables V–VII compare the different fusion technique performances for Zarmedine, Dar Chichou, and Jarvselja, respectively. The measures are done at FR for the four considered bands taken separately (i.e., RMSE) and together (i.e., ERGAS, SAM, $Q2^n$, and PSNR). As the second and third areas are highly heterogeneous due to the presence of forests, the fusion performances are generally degraded. Comparison between bands shows that the three first bands are generally better estimated than the last one. Indeed, the last is brighter due to the vegetation scattering increase, and therefore, the error increases proportionally. For Zarmedine, the error is lower than 1% for GSA and FHSSR, which is an acceptable value, as it remains lower than the sensor and atmosphere possible noise. Even though it belongs to the CS technique family that does not conserve spectral quality, GSA

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(a) (b)

Fig. 6. Original true color Sentinel-3 images. (a) Tunisia image taken at 09:44 on January 1, 2021. (b) Estonia image taken at 08:43 on June 18, 2021.



(a) (b)

(c)

Fig. 7. Original true color Sentinel-2 images. (a) Zarmedine, Monastir, Tunisia, at 10:10 on December 31, 2020. (b) Dar Chichou, Cap Bon, Tunisia, at 10:13 on December 31, 2020. (c) Jarvselja, Tartu, Estonia, at 9:40 on June 18, 2021.

produces the best results in terms of spatial (ERGAS and PSNR) and spectral (SAM) qualities. This is can be explained by the spectral closeness between the two sensors as well as the spatial homogeneity making the GSA assumption hold. For Dar Chichou, only FHSSR is always less than 1% in terms of RMSE. For this same area, GSA error increases in case B8A/Oa17 due to the heterogeneity of the scene. In this case, the model-based techniques, such as CNMF, ECCV'14, ICCV'15, and Hysure, give the best results, as they are able to deal with heterogeneous pixels. FHSSR provides

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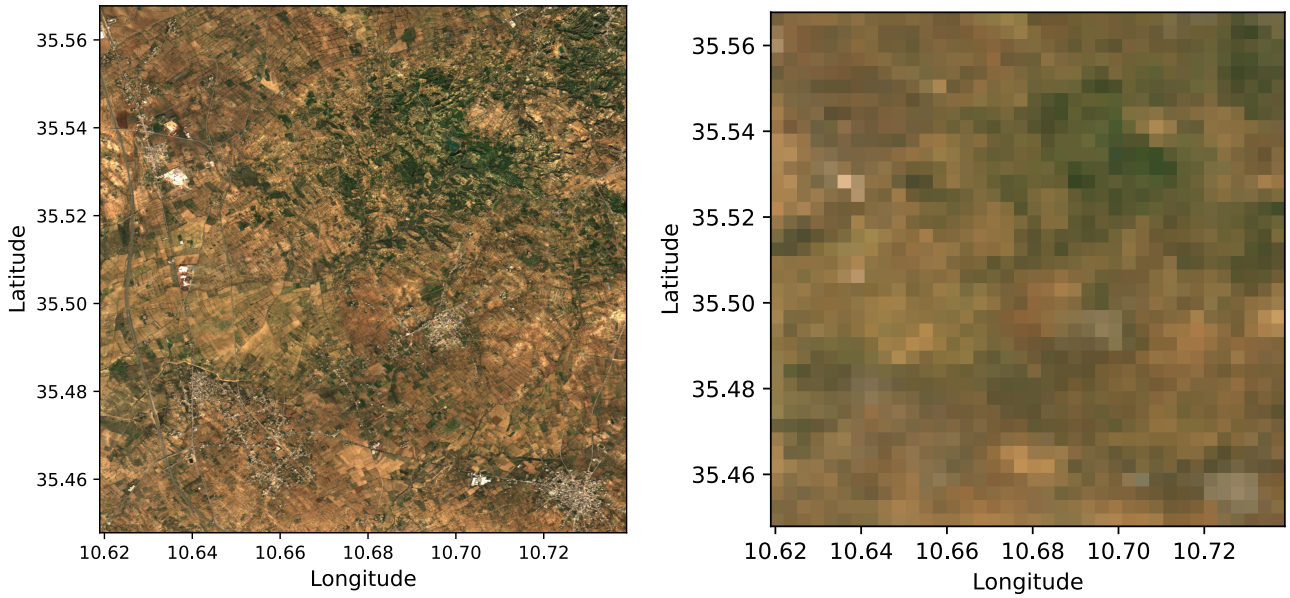


Fig. 8. Zarmedine true color at different resolutions. (a) Sentinel-2 @10 m ($\theta_s = 60.94°$ and $\theta_o = 0°$). (b) Sentinel-3 @300 m ($\theta_s = 62.69°$, $\theta_o = 6.17°$, and $\varphi = 127.6°$).

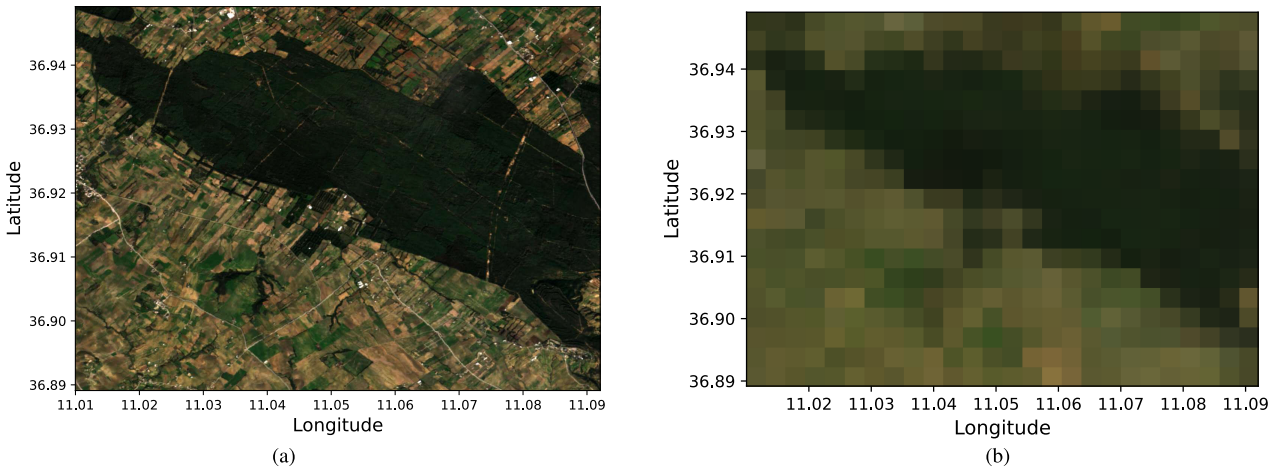


Fig. 9. Dar Chichou true color at different resolutions. (a) Sentinel-2 @10 m ($\theta_s = 61.80°$ and $\theta_o = 0°$). (b) Sentinel-3 @300 m ($\theta_s = 63.88°$, $\theta_o = 5.67°$, and $\varphi = 128°$).

better results, as it tries to find the physically based Sentinel-3 signatures given those of Sentinel-2. FHSSR0 is less effective than FHSSR, since the Sentinel-3 signatures are not physically corrected in the former case. Jarvselja's results are similar to those of Dar Chichou, since both of them correspond to heterogeneous forests. FHSSR is the best almost for all the bands followed by GSA. CNMF, ECCV'14, and HySure give acceptable results, since they are model-based allowing to deal with heterogeneity. In B8A/Oa17 case, FSSHR highly overpasses the other, since it is based on RT modeling that allows dealing with heterogeneity as well as geometry variation from Sentinel-2 to Sentinel-3. Besides, for all the areas, the traditional unmixing technique provides a high error rate, because it assumes the same reflectance of a given cluster within each block of 7 × 7 Sentinel-3 pixels (i.e., 210 × 210 pixels at full resolution). GLP and CMD belong to the MRA techniques, and thus, they produce similar results.

2) Visual Inspection: Fig. 11 shows the difference between B8A and Oa17 in the case of Dar Chichou, which is the more complicated configuration due to the high reflectance values, in addition, to being the more significant, as the spectral bands are almost the same. The unmixing technique produces the highest error level. FHSSR, ECCV'14, HySURE, and CNMF followed by ICCV'15 show the best results. In general, the error appears in heterogeneous areas and roads where high-frequency details are difficult to reconstruct.

In order to go in-depth and understand the difference between the best performing techniques (i.e., CNMR, ECCV'14, ICCV'15, HySure, and FHSSR), Fig. 12 shows a zoomed-in view of a subarea around the forest of Dar Chichou. First, it is clear that ICCV'15 colors are flat, and in general, brightness and darkness are reduced. CNMF, ECCV'14, HySure, and FHSSR are close to the original B8A band. Nevertheless, for CNMF and ECCV'14, the yellow framed

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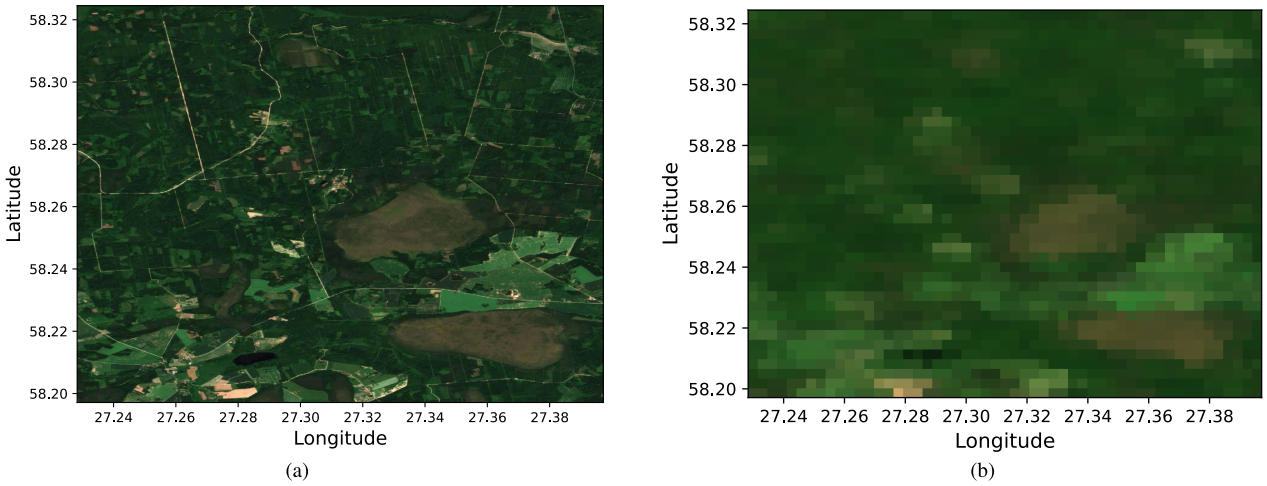


Fig. 10. Jarvselja true color at different resolutions. (a) Sentinel-2 @10 m ($\theta_s = 35°$, $\theta_o = 10°$, and $\varphi_{so} = 131°$). (b) Sentinel-3 @300 m ($\theta_s = 38.28°$, $\theta_o = 25.79°$, and $\varphi = 317.66°$).

TABLE V

QUALITY INDICES AT FULL RESOLUTION FOR ZARMEDINE AREA. FOR THE FIRST FOUR ROWS, THE METRIC IS RMSE. CYAN AND TURQUOISE COLORED CELLS CORRESPOND TO THE FIRST AND SECOND RANKED PERFORMANCE TECHNIQUES, RESPECTIVELY

Method	GSA	GLP	SFIM	CNMF	ECCV14	ICCV15	HySure	FUSE	CMD	Unmixing	FHSSR0	FHSSR
B2/Oa04	0.003	0.005	0.005	0.005	0.005	0.008	0.004	0.004	0.005	0.016	0.006	0.004
B3/Oa06	0.004	0.006	0.006	0.006	0.009	0.011	0.004	0.003	0.007	0.022	0.008	0.006
B4/Oa08	0.004	0.010	0.009	0.011	0.015	0.015	0.015	0.004	0.011	0.030	0.011	0.007
B8A/Oa17	0.005	0.016	0.015	0.009	0.009	0.009	0.008	0.013	0.018	0.029	0.025	0.004
ERGAS	0.216	0.309	0.304	0.293	0.377	0.471	0.314	0.43	0.309	0.866	0.433	0.22
SAM	2.56	2.80	2.76	3.215	4.0	3.89	3.74	5.2	2.9	5.74	5.21	2.7
Q2n	0.937	0.926	0.926	0.931	0.928	0.912	0.948	0.943	0.93	0.908	0.943	0.927
PSNR	40.59	35.49	36.15	36.11	33.59	31.89	36.83	33.44	35.61	26.74	33.43	38.13

TABLE VI

QUALITY INDICES AT FULL RESOLUTION FOR THE DAR CHICHOU AREA. FOR THE FIRST FOUR ROWS, THE METRIC IS RMSE

Method	GSA	GLP	SFIM	CNMF	ECCV14	ICCV15	HySure	FUSE	CMD	Unmixing	FHSSR0	FHSSR
B2/Oa04	0.004	0.008	0.008	0.006	0.007	0.008	0.005	0.004	0.007	0.014	0.009	0.004
B3/Oa06	0.006	0.012	0.011	0.009	0.009	0.012	0.007	0.009	0.009	0.023	0.015	0.003
B4/Oa08	0.005	0.013	0.013	0.012	0.013	0.014	0.012	0.004	0.013	0.027	0.018	0.005
B8A/Oa17	0.020	0.027	0.027	0.010	0.009	0.013	0.008	0.023	0.023	0.041	0.019	0.005
ERGAS	0.409	0.731	0.731	0.684	0.796	0.902	0.590	0.438	0.65	1.52	0.86	0.38
SAM	3.78	4.64	4.60	4.82	5.14	5.30	4.94	4.24	4.81	7.7	6.57	2.99
Q2n	0.895	0.895	0.889	0.881	0.885	0.879	0.905	0.887	0.90	0.877	0.90	0.894
PSNR	36.94	31.12	31.15	31.91	30.93	29.83	33.57	36.54	32.27	25.23	28.89	37.57

area appears darker than the original color, and for HySure, the blue framed area looks darker than the origin too.

C. Validation at Coarse Resolution

When downsampled to its original resolution (300 m), the fusion product should be as close as possible to the original Sentinel-3 image. Tables VIII–X present the quality indices taking into account the 16 Sentinel-3 bands. In general, for all the cases, the ERGAS and SAM indices are relatively low, and PSNR is high, which confirms the validity of the different techniques. CMD outperforms the others. Indeed, the different filters of this MRA technique were designed to ensure the condition that the downsampled fusion product equals the original image.

This strong constraint on the downsampled version of the image was not enough to obtain good results at FR. Indeed, the injection of the details that come from the high-resolution Sentinel-2 image is an important task, and if they are not compatible with the original low-resolution Sentinel-3 image, the fusion product will suffer from artificial artifact [see Fig. 12(i)]. CMD has shown good performances at both coarse and FRs in the original work [7], since the resolution ratio between the fused image is only 4; in addition, the images are taken with the same sensor simultaneously, and the risk of incompatibility between the approximation and details is too

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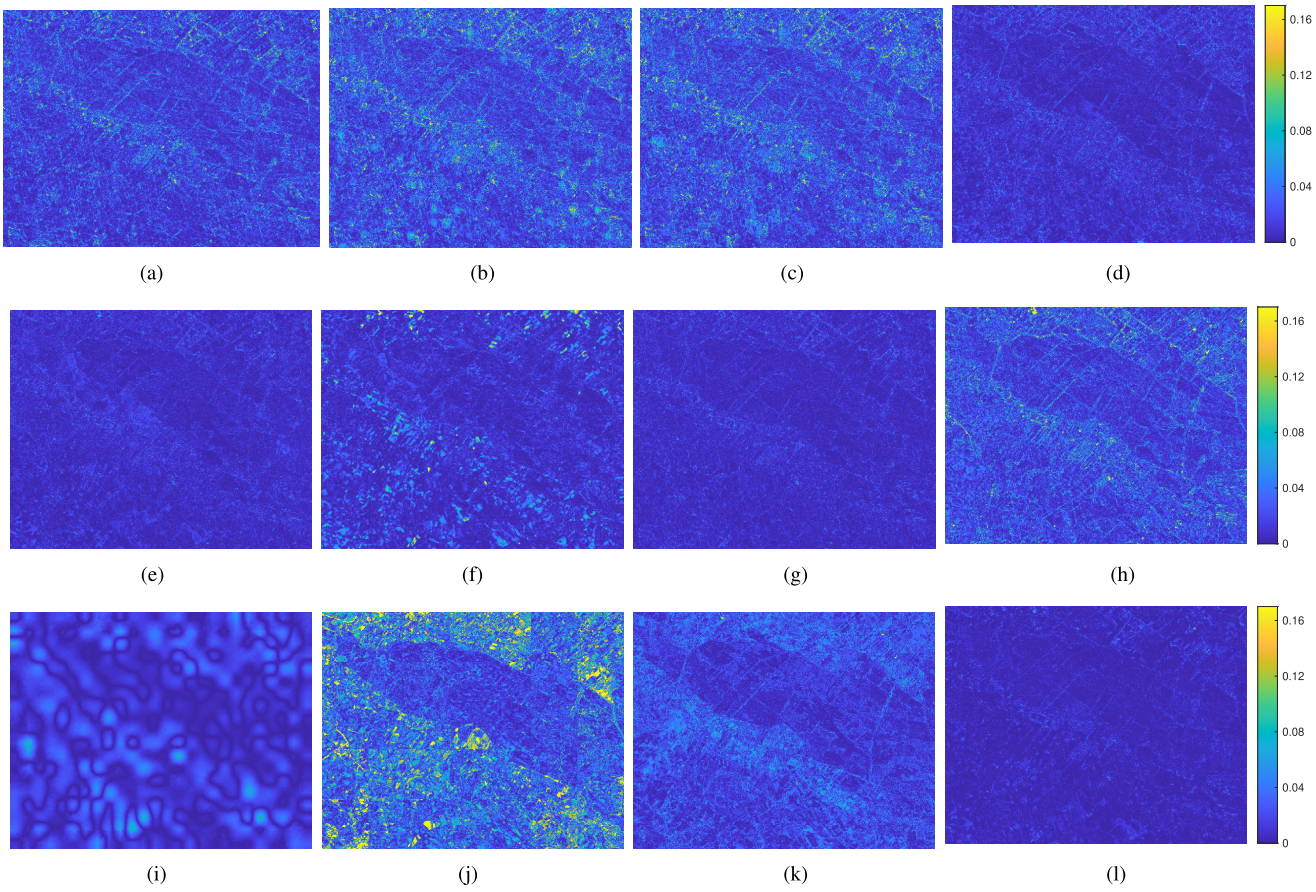


Fig. 11. Image of absolute difference between B08A and Oa17 produced at full resolution using the compared techniques for Dar Chichou. (a) GSA. (b) GLP. (c) SFIM. (d) CNMF. (e) ECCV'14. (f) ICCV'15. (g) HySure. (h) FUSE. (i) CMD. (j) Unmixing. (k) FHSSR0. (l) FHSSR.

TABLE VII
QUALITY INDICES AT FULL RESOLUTION FOR THE JARVSELJA AREA. FOR THE FIRST FOUR ROWS, THE METRIC IS RMSE

Method	GSA	GLP	SFIM	CNMF	ECCV14	ICCV15	HySure	FUSE	CMD	Unmixing	FHSSR0	FHSSR
B2/Oa04	0.006	0.006	0.006	0.006	0.005	0.006	0.005	0.006	0.006	0.008	0.005	0.004
B3/Oa06	0.009	0.010	0.010	0.010	0.011	0.011	0.010	0.009	0.010	0.015	0.012	0.003
B4/Oa08	0.001	0.005	0.005	0.004	0.005	0.005	0.004	0.005	0.005	0.010	0.007	0.003
B8A/Oa17	0.073	0.074	0.074	0.072	0.074	0.073	0.074	0.084	0.075	0.083	0.082	0.004
ERGAS	0.546	0.815	0.824	0.729	0.723	0.891	0.649	0.767	0.777	1.5	0.822	0.517
SAM	1.844	2.497	2.45	2.351	2.251	2.439	2.088	2.199	2.384	3.625	2.647	2.423
Q2n	0.764	0.555	0.558	0.774	0.719	0.891	0.714	0.7	0.663	0.473	0.453	0.612
PSNR	35.39	30.31	30.29	32.24	31.84	30.54	32.60	31.8	30.79	25.23	29.14	34.16

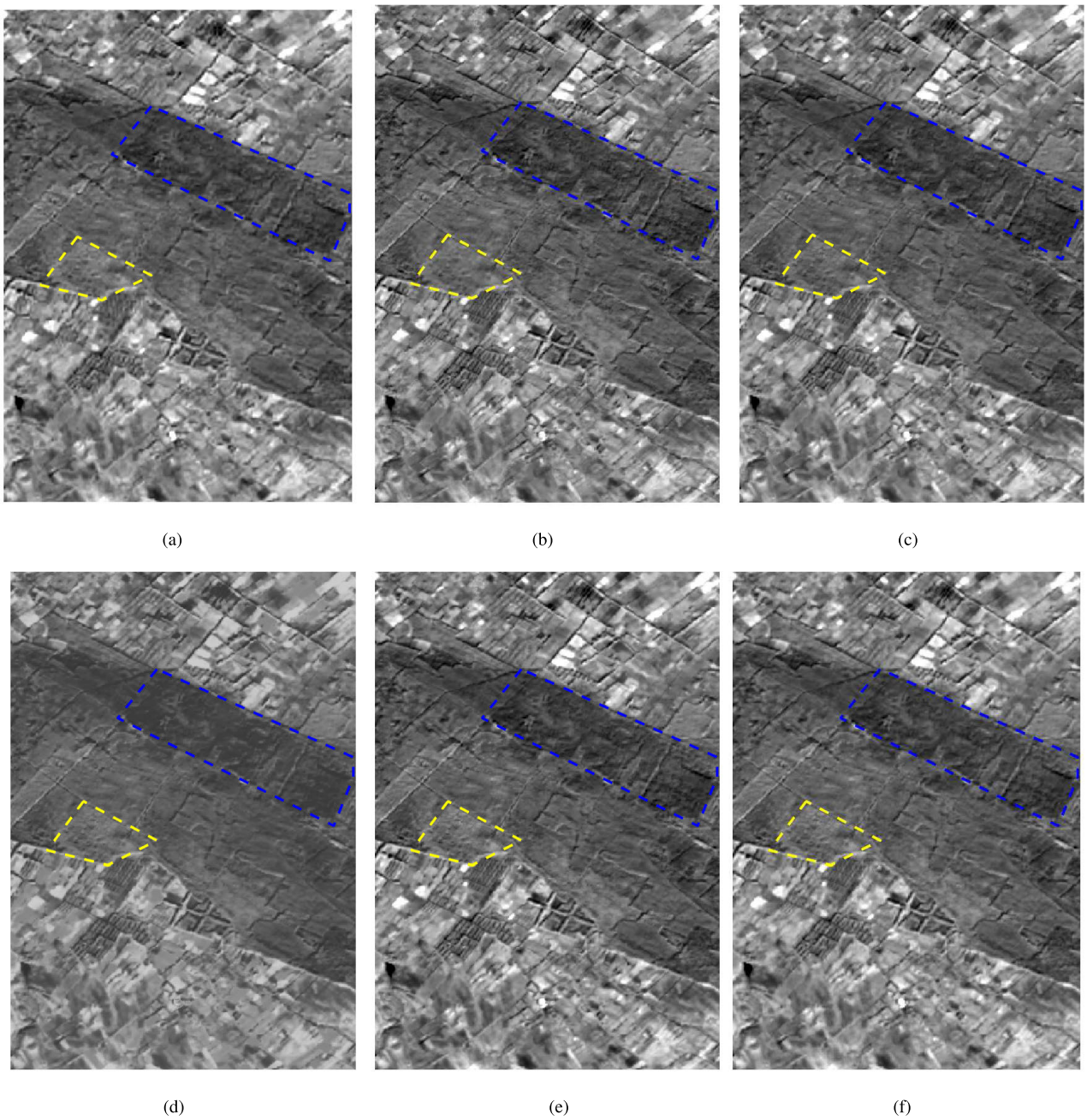
small. In contrast to CMD, the model-based technique does not force the fusion product to match at coarse resolution, but try to find a trade-off between detail injection and spectral quality preservation in order to maximize the quality of the fusion product at FR.

D. Spectra Consistency Analysis

In order to check the validity of the different fusion techniques, it is important to compare their Sentinel-3 estimated signatures with the original Sentinel-2 signature. Indeed, even though they are sampled at different wavelengths, it is possible to compare the global trend of the different signatures and see whether they are close are not to Sentinel-2.

Fig. 13 compares different spectra varying from bare soil to dense vegetation. The amount of vegetation can be roughly known according to the difference between NIR and red reflectance; a high difference corresponds to dense vegetation. For that, Fig. 13(d) shows dense vegetation. In opposite, the case of a linear increase of reflectance as a function of the wavelength corresponds, in general, to bare soil [39]. This is the case for Fig. 13(a). The comparison between the retrieved Sentinel-3 signatures and the Sentinel-2 ones shows that all the methods try to approach the Sentinel-2 ones. However, there is a great difference between the techniques in terms of accuracy. In particular, our approach, FHSSR, provides the overall best results greatly outperforming the others because of the use of an RTM, which guarantees spectral consistency of the

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(a) (b) (c)

(d) (e) (f)

Fig. 12. Comparison between B8A and Oa17 produced by the best fusion techniques for Area 2. The plots concern a zoomed-in view on a subarea around the forest. (a) Original. (b) CNMF. (c) ECCV'14. (d) ICCV'15. (e) HySure. (f) FHSSR.

TABLE VIII
QUALITY INDICES AT THE ORIGINAL RESOLUTION FOR AREA 1

Method	GSA	GLP	SFIM	CNMF	ECCV14	ICCV15	HySure	FUSE	CMD	Unmixing	FHSSR0	FHSSR
ERGAS	0.179	0.15	0.087	0.186	0.227	0.192	0.211	0.276	0.021	0.165	0.225	0.273
SAM	1.18	0.97	0.65	1.08	1.6	0.19	1.51	3.81	0.1	1.02	1.76	1.98
PSNR	30.66	32.29	37.02	30.41	28.29	30.46	29.51	28.18	49.44	31.62	28.27	27.46

results. Conversely, the unmixing technique provides the worst results, since it considers the same signature for all the pixels belonging to the same cluster and the same patch. This misfit greatly appears in the case of bare soil and dense vegetation, since they cannot be precisely represented by cluster average signatures. GSA, HySure, and FUSE, which quantitatively give the first ranked results after FHSSR, do not show good agreement with Sentinel-2. The difference is especially high in the NIR domain due to the great variance of the canopy reflectance is that spectral interval. Moreover, in addition to the bias, FUSE shows nonself consistent signatures, as they present many artifacts in their variation as the functions of the

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TABLE IX
QUALITY INDICES AT THE ORIGINAL RESOLUTION FOR AREA 2

Method	GSA	GLP	SFIM	CNMF	ECCV14	ICCV15	HySure	FUSE	CMD	Unmixing	FHSSR0	FHSSR
ERGAS	0.333	0.262	0.156	0.314	0.362	0.325	0.408	0.376	0.05	0.27	0.38	0.51
SAM	1.63	1.26	0.805	1.34	1.71	1.43	1.93	2.601	0.17	1.20	1.84	2.84
PSNR	25.76	28.03	32.63	26.16	25.09	26.16	24.44	24.48	42.0	27.54	24.13	22.79

TABLE X
QUALITY INDICES AT THE ORIGINAL RESOLUTION FOR THE JARVSELJA AREA

Method	GSA	GLP	SFIM	CNMF	ECCV14	ICCV15	HySure	FUSE	CMD	Unmixing	FHSSR0	FHSSR
ERGAS	0.405	0.293	0.15	0.354	0.479	0.423	0.472	0.518	0.075	0.293	0.38	0.759
SAM	0.954	0.692	0.446	0.687	1.53	0.857	1.86	2.618	0.13	0.661	1.84	2.355
PSNR	28.05	30.84	36.54	28.74	26.54	27.99	26.65	25.86	42.76	30.79	24.13	21.88

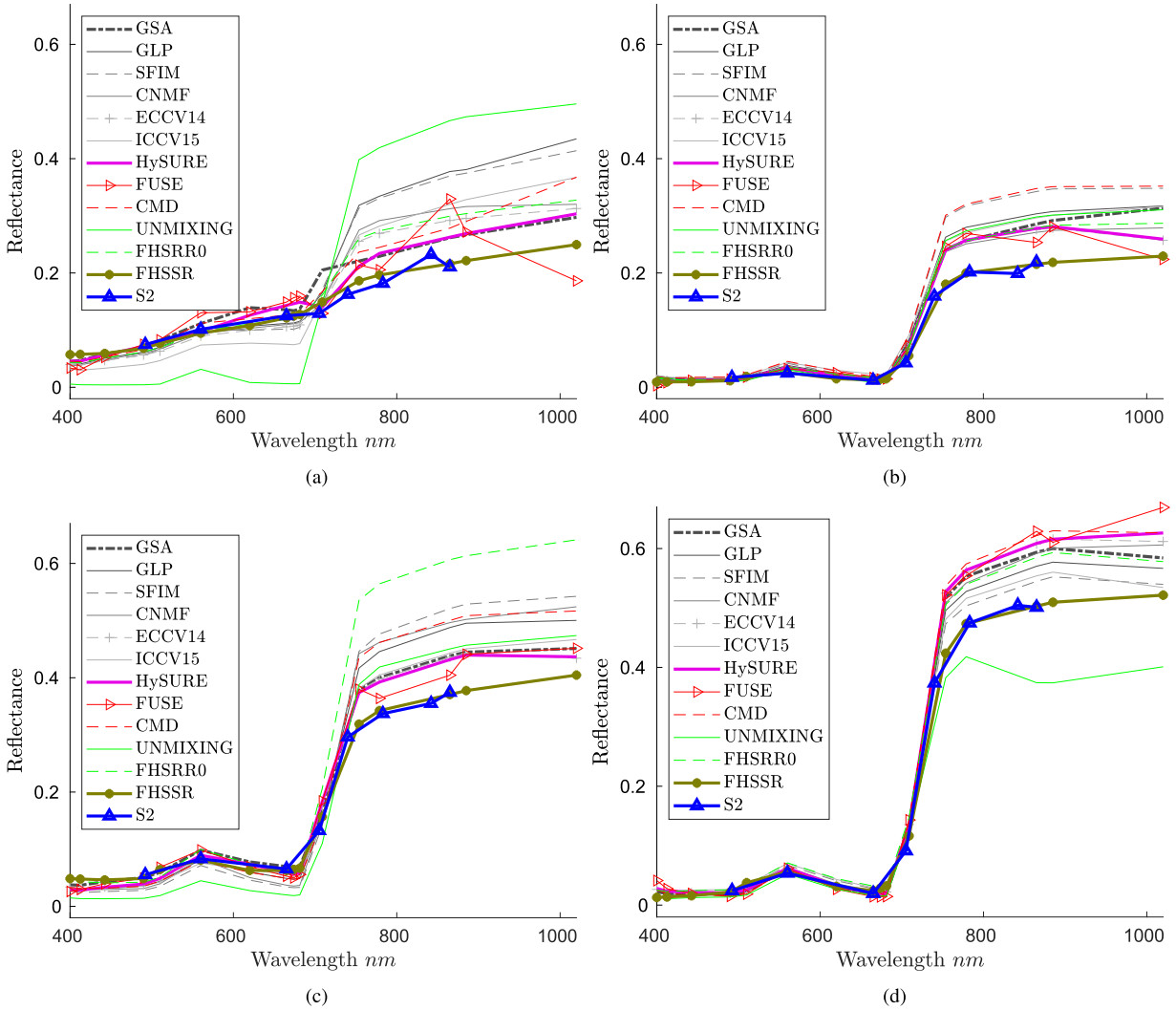


Fig. 13. Comparison of estimated Sentinel-3 signatures using the different techniques and the original Sentinel-2 image for different land-cover cases. (a) Bare soil. (b) Mixed pixel. (c) Sparse vegetation. (d) Dense vegetation.

wavelength. This is mainly the case in the NIR domain where the signatures are highly unstable. These results confirm the validity of our approach showing the effectiveness in using a physical model to link the derived Sentinel-3 signatures to the original Sentinel-2 ones in order to ensure physically consistent results.

Note, finally, that the Sentinel-2 signatures present sometimes a nonconsistency in band B8, and this is mainly because this band is relatively wide and might include several effects making the interpretation more complex than others bands.

E. Validation on a Large Test Site

Our approach has shown high performances when applied on image fusion for different kinds of vegetation densities. Nevertheless, the studied areas are relatively small and could be dominated by one or few vegetation species. This scenario

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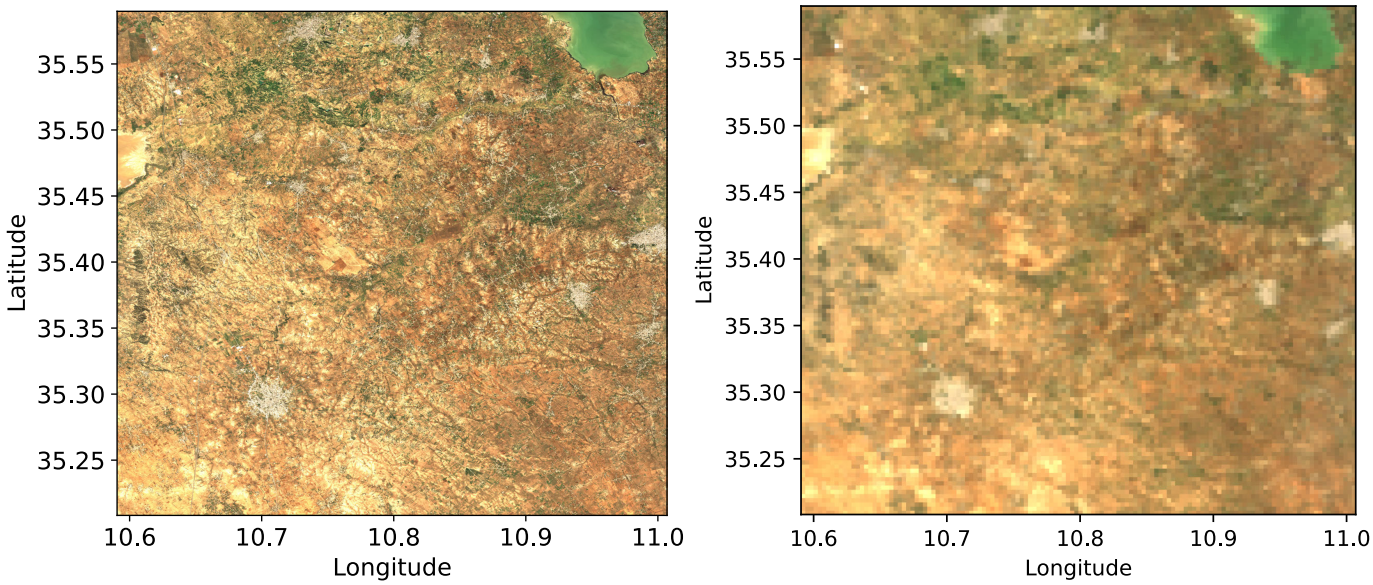


Fig. 14. True color (a) Sentinel-2 and (b) Sentinel-3 images for the Zarmedine large area.

TABLE XI

QUALITY INDICES AT FULL RESOLUTION FOR THE ZARMEDINE LARGE AREA. FOR THE FIRST FOUR ROWS, THE METRIC IS RMSE

Method	GSA	GLP	SFIM	CNMF	ECCV14	ICCV15	HySure	FUSE	CMD	Unmixing	FHSSR0	FHSSR
B2/Oa04	0.007	0.008	0.008	0.007	0.007	0.016	0.07	0.013	0.008	0.016	0.014	0.004
B3/Oa06	0.004	0.007	0.007	0.005	0.008	0.02	0.004	0.018	0.007	0.02	0.015	0.004
B4/Oa08	0.007	0.01	0.01	0.009	0.013	0.028	0.01	0.028	0.011	0.028	0.017	0.006
B8A/Oa17	0.011	0.014	0.014	0.008	0.008	0.026	0.07	0.019	0.013	0.026	0.019	0.005
ERGAS	0.286	0.368	0.363	0.33	0.44	0.822	0.328	0.711	0.36	0.822	0.817	0.221
SAM	2.48	2.69	2.63	2.77	4.16	5.24	3.55	5.22	2.77	5.25	4.48	3.12
Q2n	0.971	0.969	0.966	0.967	0.961	0.953	0.959	0.957	0.968	0.954	0.739	0.969
PSNR	46.73	43.47	43.67	44.89	41.37	36.16	45.58	37.39	43.59	36.16	36.14	47.64

TABLE XII

QUALITY INDICES AT THE ORIGINAL RESOLUTION FOR THE ZARMEDINE LARGE AREA

Method	**GSA**	**GLP**	**SFIM**	**CNMF**	**ECCV14**	**ICCV15**	**HySure**	**FUSE**	**CMD**	**Unmixing**	**FHSSR0**	**FHSSR**
ERGAS	0.173	0.135	0.076	0.169	0.306	0.16	0.238	0.109	0.044	0.16	0.388	0.304
SAM	1.137	0.844	0.584	0.956	2.283	0.91	2.151	0.609	0.257	0.91	1.94	2.045
PSNR	32.46	34.72	39.78	32.78	28.64	33.33	30.38	0.109	44.21	33.33	25.27	28.4

is representative when the focus of the study is agriculture fields or forest stands. However, in many cases, it is required to study the canopy at larger scale, such as watersheds. For this reason, in this section, we consider a larger test site in the Zarmedine area comprising heterogeneous land-cover types. Fig. 14 shows the Sentinel-2 and Sentinel-3 of this test site. Their size is 4146 × 3800 and 132 × 121 pixels, respectively.

We applied our technique with same parameters as in the previous tests, and we compared it with the other previously considered competing methods. Table XI shows the fused image results at full resolution. As for the previous tests, our approach confirms its superiority for all the bands and almost for all the metrics. When looking at the quantitative results compared with those obtained for the Zarmedine smaller area, we see globally consistent results with sometimes some improvements. For instance, the RMSE of Oa06 has decreased from 0.006 to 0.004, and the PSNR has increased from 38.13 to 47.64. This result can be explained by the fact that the larger image contains more homogeneous areas, such as the lake in the northeast part of the image. Such areas are easier to predict at high resolution. The performance amelioration is shown for almost all the techniques except FUSE that was among the best performing and then got highly degraded in this test case, suggesting that it might not be well adapted to process large heterogeneous images.

Table XII shows the performances at the original resolution. We see also almost the same performances as for the smaller area. CMD gets always the better performances, and our approach quality indices remain consistent.

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V. Conclusion

The launch of operational Earth observation satellites with different spatial, spectral, and temporal resolutions has triggered the appearance of fusion techniques trying to take advantage of the best features of the different sensors. The Copernicus satellites Sentinel-2 and Sentinel-3 are an example of satellites with complementary resolutions. Sentinel-3 is characterized by high temporal and spectral resolutions, whereas Sentinel-2 is characterized by high spatial resolution. In this study, we proposed a novel technique to fuse Sentinel-2 and Sentinel-3 images in order to obtain a product with the same spectral resolution of Sentinel-3 and at the spatial resolution of Sentinel-2.

Our approach reconstructs Sentinel-3 images at 10 m based on the decomposition of each 10-m pixel a combination of different spectral signatures. Spectra of vegetation were jointly estimated on Sentinel-2 and Sentinel-3 images. The Sentinel-3 signatures were corrected using those of Sentinel-2 by using an RTM. The pixel composition was derived using a dictionary learning technique.

The validation was done using satellite images on a rural homogeneous area and a rather heterogeneous area containing agricultural fields and forests. The validation at FR assumed that the bands Oa04, Oa06, Oa08, and Oa17 of Sentinel-3 are approximately the same as the Sentinel-2 bands B2, B3, B4, and B8A, respectively. Our approach was compared with some well-known techniques. In general, good results have been obtained, and conversely to the others, our approach performances were not degraded in heterogeneous case studies. Moreover, for the Oa17 band, which is the more difficult to predict due to its highest brittleness, our approach highly outperformed the others, since the signature is corrected using a physically based approach. The robustness of the proposed method when considering a large image size was also studied, showing consistent performances with respect to the results obtained on the other test sites, which was not always the case for the other compared techniques.

Besides, to show the physical consistency of our fusion, we compared our Sentinel-3 estimated signatures with the Sentinel-2 ones. The results show a good agreement, which is not the case for all the other techniques. Unlike the other competing techniques, our approach can take into account the scene geometry (i.e., sun and sensor angles). This advantage can be used to fuse multiangular observations at different resolutions.

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